

Mueez Mughal | Mechatronics Engineer

mueezmmughal7@gmail.com · 403-200-9308 · <https://mueezmughal.github.io/> · [linkedin.com/in/mueezmughal](https://www.linkedin.com/in/mueezmughal)

Education

University of British Columbia, Vancouver, BC

Expected May 2028

Bachelor of Applied Science - Mechanical Engineering (Mechatronics Specialization)

Dean's List

UBC Applied Science Co-op Program

Skills

Software: MATLAB, Python, C, Simulink, SolidWorks, ANSYS, KiCad, LTSpice, LabVIEW, Git, 8051 Assembly

Fabrication: CNC Milling, Lathe, SMD/THT Soldering, 3D Printing (FDM/SLA), Composite Fabrication

Mechanical: Mechanical Design, FEA, GD&T, Tolerance Stack-Up, Structural & Fatigue Analysis, Machine Design

Electronic: STM32, ESP32, Arduino, Circuit Design, PCB Design, Motor Control, Power Electronics, VHDL, Verilog, FPGA

Communication: Technical Writing, Cross-Functional Collaboration, Documentation, Leadership

Work Experience

Fingerprint Technologies, Vancouver, BC

May 2026 – Aug 2026

Mechatronics R&D Engineering Intern

- Designed a **4-layer mixed-signal PCB** controlling 2 steppers, 2 load cells, I2C ToF sensing, and a CAN HMI.
- Integrated **20 mV analog sensing** with **48 V, 11 A motor power** using noise-aware PCB layout and grounding.
- Designed a precision servo press holding **0.001 in (25.4 μ m) clearance**, cutting build time from **2–3 h to <30 min**.
- Designed an e-bike chassis validation rig to evaluate frame strength against **EN 15194 / ISO 4210** requirements.
- Performed end-to-end mechanical design for prototype and production assemblies, spanning chassis and machine levels.
- Predicted bearing press forces within **5%** using Lamé's equations, then validated loads experimentally up to **10 kN**.

Suncor Energy, Calgary, AB

May 2025 – Dec 2025

Mechanical Engineering Intern

- Developed a physics-based Python diagnostic platform detecting ESP snapped shafts **1–2 minutes before failure**.
- Analyzed **50+ ESP failure events** using motor current, power factor, and reactive power to identify failure modes.
- Modeled downhole-to-wellhead mass and energy balances to reconcile electrical input power with shaft output.
- Built induction motor equivalent circuit models to evaluate torque and efficiency under varying load conditions.
- Automated ESP lifecycle performance analytics in Python, contributing to **\$2.5M in sustained annual savings**.
- Built real-time **PI Vision dashboards** used by Firebag reservoir, production, and process engineering teams.

UBC Solar Racing, Vancouver, BC

6th Place Overall (6/24) — Qualified for American Solar Challenge

Vehicle Dynamics Lead

Sep 2025 – Present

- Lead a **13-member Vehicle Dynamics team** owning suspension, steering, and braking for UBC Solar's Gen 4 car.
- Set technical direction across **CAD, FEA, testing, validation, and manufacturing** for vehicle dynamics systems.
- Review analyses, CAD, and drawings to ensure performance, safety, and manufacturability requirements are met.
- Drive cross-functional design reviews to resolve integration issues and deliver systems ahead of design freeze.

Vehicle Dynamics Sub-Team Member

Sep 2024 – Sep 2025

Steering Wheel Design

- Designed a competition-compliant quick-release steering wheel system, reducing **driver egress time by 30%**.
- Integrated a **custom PCB** within the steering assembly with the Power and Signals team for reliable driver controls.
- Fabricated **waterjet-cut aluminum baseplates** and rapid prototypes to validate fit, ergonomics, and tolerances.

Lower Control Arm Design (Gen 4 Car)

- Designed **all 4 lower control arms** for UBC Solar's Gen 4 car, supporting the vehicle's new suspension system.
 - Performed **ANSYS FEA under 2g bump, 1g cornering, and 1g braking** to validate static and fatigue strength.
 - Conducted destructive testing on suspension rod ends to determine yield strength and characterize failure modes.
 - Produced manufacturing drawings with **GD&T** to control tolerance stack-up and ensure repeatable fabrication.
 - Developed a **2-DOF quarter-car MATLAB model** to identify ride frequencies and validate spring-damper selection.
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Projects

Homemade Capacitive Weight Sensor

Nov 2024

- Modeled capacitance ($C = \kappa A/d$) versus applied load using solid mechanics of polyurethane foam deformation.
- Designed and built a parallel-plate capacitive weight sensor capable of measuring load with **~99% accuracy**.